

Practice Quiz: Perception

Part A: Multiple Choice

1. Yarn tension felt through the crocheter's fingers is best described in Active Inference as:
A) An internal state of the completed fabric B) A sensory state carrying information across the Markov blanket C) An external state unrelated to the crocheter D) A planning variable for future projects
2. When a crocheter detects a dropped stitch by the change in how the fabric feels, this is an example of: A) Free energy minimization through model updating B) Surprise — sensory evidence deviating from the generative model's prediction C) The Markov blanket dissolving D) External states becoming more organized
3. An experienced crocheter can identify yarn fiber type by touch alone. This ability reflects: A) Low precision in tactile processing B) A highly refined generative model that generates precise predictions about different fibers C) No sensory processing at all — it is purely guessing D) The crocheter ignoring tactile information
4. When crocheting in dim light, a crocheter tends to rely more on touch. In Active Inference terms, this is because: A) Visual information becomes less reliable (lower precision), so tactile channels are weighted more heavily B) The crocheter has lost their generative model C) Touch is always more precise than vision D) Dim light increases free energy in all channels equally
5. A crocheter describes being "in the rhythm" during repetitive single crochet rows. This rhythm reflects: A) The absence of a generative model B) The generative model's temporal predictions matching the observed tactile pattern C) Maximum free energy D) The crocheter not paying attention at all
6. When a hook splits the yarn (catches only some plies), the crocheter immediately feels something is wrong. This is because: A) The tactile feedback matches the expected resistance perfectly B) The prediction error is zero C) The resistance and texture of the

pull-through deviate from the predicted sensation D) Split yarn always looks different but feels the same

7. A beginner crocheter must consciously attend to finger tension, while an expert maintains even tension automatically. This transition illustrates: A) The expert losing their perceptual abilities B) Perceptual inference becoming habituated as the generative model gains precision C) The beginner having a better generative model D) Tension being irrelevant to perception

Part B: Short Answer

1. Describe the tactile difference between working a single crochet and a double crochet from the crocheter's perspective. How does the generative model predict different resistance patterns for each stitch type?
2. Explain why working with an unfamiliar yarn fiber (one you have never used before) tends to slow you down and require more conscious attention, using the concepts of precision and prediction error.
3. A crocheter is working in a noisy, distracting environment. Which perceptual channels might suffer, and how might the crocheter compensate by adjusting precision weighting across sensory modalities?